

STRIVE

Mathematical Specification

Every formula in the system, and why it was chosen

AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

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Scope of claims. This document specifies the mathematics that the implementation actually computes. Where a quantity is a heuristic, a proxy, or uncalibrated, it is labelled as such. No performance claim appears here that was not measured; measured values cite the evidence file that contains them.

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1 Notation and conventions

Let $x[n] \in \mathbb{R}$ denote a discrete-time audio signal at sampling rate $f_s = 16,000$ Hz, normalised so $x[n] \in [-1, 1]$. Time in seconds is $t = n/f_s$.

Symbol	Meaning
f_s	Sampling rate, fixed at 16,000 Hz
W	Analysis window length in seconds (default 2.0)
H	Hop length in seconds (default 1.0, supports 0.5)
$N_W = W f_s$	Window length in samples
$N_H = H f_s$	Hop length in samples
$\hat{v}$	Unit-normalised vector, $\hat{v} = v / \max(\ v\ _2, \varepsilon)$
s_{glob}	Global retrieval track score (artifact evidence)
s_{sess}	Session consistency track score
s_{coh}	Boundary coherence track score
R_a	Authenticity risk (acoustic, temporally smoothed)
R_c	Context risk (business metadata)
R_d	Decision risk (policy combination)
ε	Numerical floor, 10^{-12} unless stated

Why this choice: unit normalisation everywhere

All embeddings are L^2 -normalised before comparison. On the unit sphere the inner product equals the cosine similarity, $\langle \hat{u}, \hat{v} \rangle = \cos \theta$, so a single FAISS inner-product index computes cosine similarity exactly with no extra work. It also removes loudness as a confound: a quiet and a loud recording of the same spectral shape map to the same point.

2 Signal conditioning

2.1 Canonical form and resampling

Every input is reduced to mono by channel averaging, $x[n] = \frac{1}{C} \sum_{c=1}^C x_c[n]$, then resampled to f_s .

Resampling uses a polyphase rational filter. For input rate r and target f_s , let $g = \gcd(r, f_s)$ and set the interpolation and decimation factors

$$L = \frac{f_s}{g}, \quad M = \frac{r}{g}. \quad (1)$$

The signal is upsampled by L (zero insertion), low-pass filtered at $\omega_c = \pi / \max(L, M)$, and downsampled by M .

Why this choice: polyphase over naive interpolation

Rational L/M resampling with an anti-aliasing filter is exact for the common telephony rates (8 kHz $\rightarrow$ 16 kHz is $L=2, M=1$; 48 kHz is $L=1, M=3$; 44.1 kHz is $L=160, M=441$). Naive linear interpolation would alias high-frequency energy into the band where synthesis artifacts live — precisely the evidence the detector depends on. Corrupting that band to save compute would be self-defeating.

2.2 Overlapping window extraction

Windows are extracted at a fixed hop. Window k spans samples

$$\mathcal{W}_k = \{ kN_H, kN_H + 1, \dots, kN_H + N_W - 1 \}, \quad k = 0, 1, 2, \dots \quad (2)$$

with start time $t_k = kH$ and end time $t_k + W$. The overlap between consecutive windows is $W - H$ seconds; the *fresh* (previously unseen) portion is the last H seconds, except for $k = 0$ where the whole window is fresh.

The constraint $0 < H \leq W$ is enforced, and both N_W and N_H must be integers. For $W = 2, H = 0.5$ the emission rate is $1/H = 2$ windows per second.

Why this choice: fresh-sample accounting

Voiced-time accumulation uses only the fresh portion: $T_{\text{voiced}} += (|\text{fresh}|/f_s) \cdot a$ where a is the activity fraction. Using the whole window would count overlapped audio W/H times, inflating the evidence counter by $4\times$ at a 0.5 s hop and letting the system declare sufficient evidence long before it exists.

2.3 Energy activity gate

The signal is framed into 20 ms blocks (320 samples). For block m ,

$$\rho_m = \sqrt{\frac{1}{320} \sum_{n \in \text{block } m} x[n]^2}, \quad a = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[\rho_m > \tau_a], \quad \tau_a = 0.008. \quad (3)$$

A window is analysed when $a \geq 0.25$.

Limitation: this is an energy detector, not a speech classifier

a measures whether *anything* is above the noise floor. Measured on 40 matched ASVspoof clip pairs, additive noise at 10 dB SNR raises the mean voiced fraction from 0.564 to 1.000 — a $1.77\times$ inflation in which every frame is counted as speech. Consequently noisy clips clear the evidence gate $5.6\times$ more often than clean ones (28 vs. 5 clips) while their discriminative power is at its worst. Research mode substitutes a trained WebRTC VAD. See §10.

3 Spectral representations

3.1 Short-time Fourier transform

With analysis window $w[\cdot]$ of length $N = 400$ (25 ms), hop 160 (10 ms), and FFT size 512,

$$X[m, \kappa] = \sum_{n=0}^{N-1} x[mR + n] w[n] e^{-j2\pi\kappa n/512}, \quad P[m, \kappa] = |X[m, \kappa]|^2. \quad (4)$$

Bin κ corresponds to frequency $f_\kappa = \kappa f_s / 512 = 31.25\kappa$ Hz.

Why this choice: 25 ms / 10 ms framing

Speech is approximately stationary over 20–30 ms. Shorter frames lose frequency resolution ($\Delta f = f_s/512 = 31.25$ Hz here, enough to resolve the harmonic comb of a typical F_0); longer frames smear transients, which is where vocoder artifacts concentrate. The 10 ms hop gives $2.5\times$ oversampling so a transient cannot fall between frames.

3.2 Countermeasure embedding (demo extractor)

The power spectrogram is pooled into $B = 32$ contiguous frequency bands. For band b ,

$$\beta[m, b] = \frac{1}{|b|} \sum_{\kappa \in b} P[m, \kappa], \quad \tilde{\beta}[m, b] = \sqrt{\frac{\beta[m, b]}{\sum_{b'} \beta[m, b']}}. \quad (5)$$

The square root of the normalised band energy is the *Hellinger* representation: treating $\beta[m, \cdot] / \sum \beta$ as a probability distribution over bands, $\|\tilde{\beta}_1 - \tilde{\beta}_2\|_2$ is (up to $\sqrt{2}$) the Hellinger distance between spectra.

The embedding concatenates the temporal mean and standard deviation:

$$v_{\text{cm}} = [\mathbb{E}_m[\tilde{\beta}[m, \cdot]] \parallel \widehat{\sigma}_m[\tilde{\beta}[m, \cdot]]] \in \mathbb{R}^{64}. \quad (6)$$

Why this choice: mean and variance, not mean alone

The mean captures the average spectral envelope; the standard deviation captures how much that envelope *moves*. Synthetic speech is frequently over-smooth in time — a vocoder produces less frame-to-frame spectral variance than a physical vocal tract. The σ half of the vector is what lets a purely unsupervised representation carry any spoof-relevant signal at all.

Limitation: this is a DSP surrogate

v_{cm} is hand-designed, not learned. It is an engineering stand-in so the streaming architecture can be built and measured before neural weights are available. Measured AUC on real ASVspoof 2019 LA data is 0.707 (§10) — weak but non-trivial signal, and explicitly the floor a trained countermeasure must beat. Every event it produces carries `demo_only: true`.

3.3 Mel filterbank and cepstral coefficients

The mel scale approximates human pitch perception:

$$\text{mel}(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right), \quad \text{mel}^{-1}(\mu) = 700 \left(10^{\mu/2595} - 1 \right). \quad (7)$$

Centre frequencies $\{h_i\}_{i=0}^{27}$ are placed uniformly in mel and mapped back to Hz, giving 26 triangular filters

$$\Lambda_i(f) = \max \left(0, \min \left(\frac{f - h_i}{h_{i+1} - h_i}, \frac{h_{i+2} - f}{h_{i+2} - h_{i+1}} \right) \right). \quad (8)$$

MFCCs are the DCT-II of the log filterbank energies, retaining the first 13:

$$c_j = \sum_{i=0}^{25} \log(\max(E_i, \varepsilon)) \cos \left[\frac{\pi j (i + \frac{1}{2})}{26} \right], \quad j = 0, \dots, 12. \quad (9)$$

Why this choice: log then DCT

Source-filter theory models speech as excitation convolved with a vocal-tract response. Convolution becomes multiplication in the spectral domain and *addition* after the logarithm, so the log-spectrum separates the two additively. The DCT then approximately decorrelates the result (it is the KLT for a first-order Markov process) and compacts energy into low indices, so 13 coefficients retain most of the vocal-tract information. Truncating discards the fine harmonic structure, which is dominated by F_0 and by speaker identity rather than synthesis artifacts.

3.4 Spectral shape descriptors

Per frame, with $\tilde{P}[\kappa] = P[\kappa] + \varepsilon$:

$$\text{centroid: } \mu_f = \frac{\sum_{\kappa} f_{\kappa} \tilde{P}[\kappa]}{\sum_{\kappa} \tilde{P}[\kappa]} \quad \text{spectral “centre of mass”} \quad (10)$$

$$\text{rolloff: } f_{85} = \min \left\{ f_{\kappa} : \sum_{\kappa' \leq \kappa} \tilde{P}[\kappa'] \geq 0.85 \sum_{\kappa'} \tilde{P}[\kappa'] \right\} \quad \text{where the energy ends} \quad (11)$$

$$\text{flatness: } \Phi = \frac{\exp\left(\frac{1}{K} \sum_{\kappa} \ln \tilde{P}[\kappa]\right)}{\frac{1}{K} \sum_{\kappa} \tilde{P}[\kappa]} \quad \text{geometric / arithmetic mean} \quad (12)$$

Why this choice: flatness as a tonality measure

By the AM–GM inequality $\Phi \in (0, 1]$, with $\Phi = 1$ exactly when the spectrum is flat (white noise) and $\Phi \rightarrow 0$ for a pure tone. It is therefore a scale-invariant tonality index. Synthesis systems often mismodel the noise/harmonic balance of fricatives and breathiness, which Φ summarises in one number.

3.5 Pitch, jitter and shimmer

Fundamental frequency is estimated by autocorrelation. For a mean-removed frame f ,

$$r[\ell] = \sum_n f[n] f[n + \ell], \quad \ell^* = \arg \max_{40 \leq \ell \leq 320} r[\ell], \quad F_0 = \frac{f_s}{\ell^*}. \quad (13)$$

The search range $\ell \in [40, 320]$ corresponds to $F_0 \in [50, 400]$ Hz. The normalised peak $r[\ell^*]/r[0]$ is the harmonicity; frames below 0.25 (or 0.5 for the cycle analysis) are treated as unvoiced and excluded.

With consecutive glottal period lengths $T_1, \dots, T_K$ and peak-to-peak cycle amplitudes $A_1, \dots, A_K$:

$$\text{jitter} = \frac{\frac{1}{K-1} \sum_{i=1}^{K-1} |T_{i+1} - T_i|}{\frac{1}{K} \sum_i T_i}, \quad \text{shimmer} = \frac{\frac{1}{K-1} \sum_{i=1}^{K-1} |A_{i+1} - A_i|}{\frac{1}{K} \sum_i A_i}. \quad (14)$$

Why this choice: jitter and shimmer as physiological cues

Human phonation is driven by tissue with mass and elasticity, so consecutive glottal cycles are never identical: healthy voices exhibit small but non-zero period and amplitude perturbation. Many synthesis pipelines generate excitation from a smooth parametric F_0 contour and therefore produce jitter and shimmer that are *too low*. These are dimensionless ratios, so they are invariant to gain and to the speaker’s absolute pitch.

Limitation: weak, confoundable cues

Jitter and shimmer are sensitive to background noise, codec quantisation and the voicing decision itself. They are supporting evidence only and are never used as a standalone spoofing verdict.

4 Channel intelligence

Channel measurements describe the *transport*, never the speaker. A degraded channel means our evidence deserves less trust; it is never itself evidence of a clone. Every quantity below is deterministic and model-free. Unavailable quantities are returned as `null`, never as 0.

4.1 Clipping ratio and level

$$\text{clip} = \frac{1}{N} \sum_n \mathbf{1}[|x[n]| \geq 0.99], \quad L_{\text{dBFS}} = 20 \log_{10} \sqrt{\frac{1}{N} \sum_n x[n]^2}. \quad (15)$$

L_{dBFS} is null when the RMS is below 10^{-9} (digital silence), since $\log_{10} 0$ is undefined and reporting 0 dBFS would falsely indicate a full-scale signal.

4.2 Occupied bandwidth

Let $\bar{P}[\kappa]$ be the time-averaged power spectrum. The occupied bandwidth is the 99% cumulative-power rolloff:

$$B = f_{\kappa^*}, \quad \kappa^* = \min \left\{ \kappa : \sum_{\kappa' \leq \kappa} \bar{P}[\kappa'] \geq 0.99 \sum_{\kappa'} \bar{P}[\kappa'] \right\}. \quad (16)$$

Why this choice: cumulative rolloff over a -3 dB cutoff

A -3 dB cutoff must be measured relative to a passband reference, which is unstable for speech because the passband level varies with phonation. The cumulative-energy definition needs no reference level and degrades gracefully: it always returns a frequency, and the fraction (0.99) sets exactly how much tail energy is ignored. Measured on the codec matrix it cleanly separates narrowband transport (≈ 2672 Hz for G.711 and 8 kHz resampling) from clean wideband (≈ 3625 Hz).

Limitation: content dependence

B depends on what is being said, not only on the channel. A low-pitched or quiet speaker on a wideband line can read as narrowband. B is a channel *hint*, not a codec classifier.

4.3 Signal-to-noise ratio proxy

Noise is estimated per frequency bin by *minimum statistics*: over the frame index m , the low quantile of the power in each bin approximates the stationary noise floor, because speech is intermittent while noise is persistent.

$$\hat{n}[\kappa] = Q_{0.10}(\{P[m, \kappa]\}_m), \quad \hat{N} = \sum_{\kappa} \hat{n}[\kappa], \quad \hat{S} = \sum_{\kappa} \mathbb{E}_m[P[m, \kappa]] - \hat{N}, \quad (17)$$

$$\text{SNR}_{\text{dB}} = \text{clip} \left(10 \log_{10} \frac{\hat{S}}{\max(\hat{N}, 10^{-6}\hat{T})}, 0, 60 \right), \quad \hat{T} = \hat{S} + \hat{N}. \quad (18)$$

Why this choice: per-bin minimum statistics, not frame energy percentiles

The first implementation used the ratio of the 95th to the 10th percentile of *frame energy*. That collapses on stationary input: for a signal with constant energy every frame is identical, the ratio is 1, and the reported SNR is 0 dB regardless of how clean the signal is. In testing, a clean tone scored 0.0 dB while the same tone plus noise scored 0.6 dB — the metric was inverted. Operating per frequency bin fixes this, because even a stationary tone occupies few bins and leaves the rest at the true noise floor. The floor $10^{-6}\hat{T}$ prevents a division blow-up on digitally clean input, which saturates to the 60 dB cap instead of returning “unmeasurable”.

Limitation: a proxy, not a measurement

Verified monotonic in added noise (true 0/5/10/20/40 dB $\rightarrow$ estimated 11.8/15.1/19.5/28.7/47.9 dB) but biased high by roughly 8 dB, because the 10th percentile underestimates the floor. Use it for ordering and for reliability weighting; do not quote it as a calibrated SNR.

4.4 Cross-window discontinuity

Let $\phi_k = (\rho_k, \mu_{f,k}, \nu_k, e_k)$ summarise fresh audio in window k : RMS, spectral centroid, noise floor (10th percentile frame energy) and final sample. Define a scale-free log jump

$$J(u, v; \delta) = \text{clip}\left(\frac{|\ln \frac{u+\delta}{v+\delta}|}{\ln 10}, 0, 1\right), \quad (19)$$

which reaches 1 at one order of magnitude of change. The discontinuity score is

$$D_k = \max\left\{J(\rho_k, \rho_{k-1}; 10^{-3}), J(\nu_k, \nu_{k-1}; 10^{-8}), \frac{|\mu_{f,k} - \mu_{f,k-1}|}{f_s/4}, \frac{|x_k[0] - e_{k-1}|}{\max(4\rho_k, 10^{-6})}\right\} \quad (20)$$

with $D_0 = 0$ (no predecessor).

Why this choice: maximum rather than sum

A splice need only disturb *one* of these quantities: an edit can preserve energy while shifting the noise floor, or preserve spectrum while leaving a sample-level step. A sum would dilute a single strong indicator across four terms; the maximum preserves it. The additive offsets δ keep the log finite when either side is silent.

Limitation: packet loss is invisible here

Measured on the codec matrix, 1%, 3% and 5% packet loss produce channel readings identical to clean audio. Twenty-millisecond zero-erasures do not move the per-window aggregate energy, centroid or noise floor enough to register. Consequently `packet_loss_rate` must be supplied by the transport and is never inferred from the waveform. Detecting erasures from audio requires a dedicated zero-run detector, not a refinement of D_k .

4.5 Reliability aggregation

Channel quality is one minus a sum of bounded penalties:

$$Q = \text{clip}\left(1 - \sum_i w_i \pi_i, 0, 1\right), \quad (21)$$

$$\pi_{\text{clip}} = \min\left(1, \frac{\text{clip}}{0.02}\right), \quad \pi_{\text{disc}} = D_k, \quad (22)$$

$$\pi_{\text{snr}} = \text{clip}\left(\frac{25 - \text{SNR}_{\text{dB}}}{25 - 5}, 0, 1\right), \quad \pi_{\text{bw}} = \text{clip}\left(\frac{7000 - B}{7000 - 3400}, 0, 1\right), \quad (23)$$

with default weights $w = (0.25, 0.20, 0.30, 0.25)$. A penalty whose input is unavailable is *omitted*, not set to zero, and Q itself is null when the window contains no usable audio.

Why this choice: linear ramps with explicit anchors

Each π_i is a clipped linear ramp between a “good” and a “bad” anchor (25 → 5 dB SNR; 7000 → 3400 Hz bandwidth, the latter being the G.711 passband). Linear ramps are monotone, bounded, and every constant is a physically interpretable number that a reviewer can argue with — unlike a fitted sigmoid, which would imply calibration we have not earned. All constants live in configuration.

Q is reliability, not risk

Q answers “how much should we trust the acoustic evidence from this window?” It never answers “is this a clone?”. A pristine channel does not make a caller genuine, and a bad channel does not make a caller fake. Keeping these separate is why Q is consumed by fusion weighting and never added to the risk score.

5 Evidence tracks

5.1 Track 1 — labelled-reference retrieval

The reference index holds unit embeddings $\{\hat{u}_i\}_{i=1}^M$ with labels $y_i \in \{0, 1\}$ (0 bonafide, 1 spoof). For a query $\hat{q}$, retrieve the k nearest neighbours by inner product, $\mathcal{N}_k(\hat{q}) = \arg \text{top-}k_i \langle \hat{q}, \hat{u}_i \rangle$, and score

$$s_{\text{glob}} = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\hat{q})} y_i \in [0, 1], \quad (24)$$

the spoof fraction of the neighbourhood. The index is partitioned by language, falling back to a global pool when a partition has fewer than 20 entries or contains a single class; the fallback is reported in the event as a reason code.

Why this choice: k -NN retrieval over a trained classifier head

A retrieval score is interpretable and auditable: every decision can be traced to the specific labelled clips that produced it, which matters for a fraud-prevention tool that must justify holding a transaction. It also extends to new attack generators by adding data rather than retraining. The cost is that quality is bounded by the corpus: a single-class or tiny partition cannot produce a meaningful score, which is exactly why the guard returns `null` instead of a number.

5.2 Track 2 — session consistency

Within a call, a bounded store of accepted profile vectors is maintained. The similarity of the current window to that store is

$$\sigma = \underbrace{\frac{1}{2} \left(\frac{1}{k} \sum_{i \in \mathcal{N}_k} \langle \hat{p}, \hat{p}_i \rangle \right)}_{\text{speaker-level profile}} + \underbrace{\frac{1}{2} \left(\frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \frac{1}{k} \sum_{j \in \mathcal{N}_k(s)} \langle \hat{s}, \hat{s}_j \rangle \right)}_{\text{segment-level}}, \quad s_{\text{sess}} = \text{clip}(1 - \sigma, 0, 1). \quad (25)$$

Trust-gated admission. A window updates the store only if

$$s_{\text{glob}} < 0.30 \quad \text{and} \quad \sigma > 0.70. \quad (26)$$

Bootstrap requires *all* initial window scores below 0.30, at least 3 valid windows, and ≥ 4 s of voiced audio by the 5 s mark; otherwise the store is permanently blocked for that call.

Why this choice: a dual gate against profile poisoning

The session profile answers “is the voice still the same as it was?” — which is only meaningful if the initial voice was genuine. Without gating, an attacker who is cloned from the first second would have the clone admitted as the reference, and every subsequent cloned window would score as perfectly consistent. Requiring low global risk *and* high self-similarity means an admitted window must look genuine to the independent retrieval track and be consistent with what is already stored.

Limitation: not proof against slow poisoning

The gate stops obvious contamination. A patient attacker who drifts the voice slowly, keeping each step inside both thresholds, is not addressed by this mechanism and has not been evaluated.

5.3 Track 3 — boundary coherence

The seam between old and new audio lies at $\tau = W - H$ seconds inside the current window. Among adjacent segment pairs (a, b) bracketing τ with a gap $b_{\text{start}} - a_{\text{end}} \leq 0.12\text{s}$, select the pair whose midpoint is closest to τ , and require that midpoint to be within 0.2s of τ . Then

$$s_{\text{coh}} = \text{clip}(1 - \langle \hat{v}_a, \hat{v}_b \rangle, 0, 1). \quad (27)$$

If no admissible pair exists, the track returns `null`.

Why this choice: clipping a quantity whose natural range is $[0, 2]$

Cosine similarity lies in $[-1, 1]$, so $1 - \cos$ lies in $[0, 2]$. Values above 1 mean anti-correlated embeddings, which for spectral segments indicates a pathological comparison rather than a stronger splice. Clipping keeps all track scores on a common $[0, 1]$ scale so the fusion weights mean the same thing for every track.

Limitation: a weak and confoundable cue

Adjacent *natural* phonemes legitimately have different embeddings — that is what makes them different phonemes. High s_{coh} therefore does not imply splicing, and the cue is further confounded by packet loss and phonetic content. It is supporting evidence with a small weight and has not been validated as a discriminator.

6 Fusion and temporal smoothing**6.1 Age-scheduled weights**

Let A be the call age in seconds. The nominal weight vector over $(s_{\text{glob}}, s_{\text{sess}}, s_{\text{coh}})$ is

$$w(A) = \begin{cases} (0.80, 0.00, 0.20) & A < 15 \\ (0.50, 0.30, 0.20) & 15 \leq A \leq 60 \\ (0.30, 0.50, 0.20) & A > 60. \end{cases} \quad (28)$$

Why this choice: shifting weight from global to session evidence

Early in a call there is no session history, so the session weight is exactly zero — not a small number, zero. As the call proceeds an in-call profile accumulates and becomes informative about mid-call speaker substitution, which the global track cannot see. The schedule encodes that changing information content.

6.2 Availability masking

Let $m_i = \mathbf{1}[s_i \neq \text{null}]$. The effective weights are

$$\tilde{w}_i = \frac{m_i w_i}{\sum_j m_j w_j} \quad \text{if} \quad \sum_j m_j w_j > 0, \quad (29)$$

and the raw fused score is $s_{\text{raw}} = \sum_i \tilde{w}_i s_i$. If s_{glob} is unavailable, *no* risk is published.

Absent evidence is not evidence of innocence

The single most important line in the fusion is that a missing track is masked and the remaining weights renormalised — it does *not* contribute $s_i = 0$. Substituting zero would let a failed model, a cold-start session profile, or a disabled identity branch actively push the risk *down*, manufacturing confidence from silence. The same principle makes a stale scheduled branch unavailable rather than reusing an expired value indefinitely.

6.3 Exponential moving average

$$R_a^{(k)} = \alpha R_a^{(k-1)} + (1 - \alpha) s_{\text{raw}}^{(k)}, \quad \alpha = 0.7, \quad R_a^{(-1)} = 0. \quad (30)$$

This is a first-order IIR low-pass filter with impulse response $(1 - \alpha)\alpha^j$, effective memory $\sum_j j(1 - \alpha)\alpha^j = \alpha/(1 - \alpha) \approx 2.33$ windows, and -3 dB cutoff at $\omega_c = \arccos(2 - \frac{1+\alpha^2}{2\alpha})$.

Starting from zero, k consecutive maximal inputs give

$$R_a^{(k)} = 1 - \alpha^{k+1}. \quad (31)$$

Why this choice: smoothing is a false-positive control

A single anomalous window — a cough, a codec glitch, one bad retrieval — must not trigger a transaction hold. Requiring sustained evidence is the mechanism that makes “verify, do not block” workable in production.

Limitation: documented counterexamples, deliberately preserved

Two consequences are pinned by unit tests rather than tuned away:

1. **Warm-up.** Three maximal updates give $1 - 0.7^3 = 0.657 < 0.75$, so an alert cannot fire in three windows; five give $1 - 0.7^5 \approx 0.832$. The source design document’s “alert in three chunks” claim is arithmetically wrong.
2. **Mature-call dilution.** For $A > 60$ with all tracks available, $s = (1, 0, 0)$ fuses to 0.30 — below the warning band. A static clone that matches its own bootstrap profile can therefore suppress a maximal global alarm. This is a real weakness of the specified schedule and is recorded, not hidden.

7 Risk separation and policy

7.1 Context risk

Business metadata maps to a bounded score by additive contributions:

$$R_c = \min \left(1, \underbrace{\begin{cases} 0.35 & \text{amount} \geq 10^6 \\ 0.15 & \text{amount} \geq 10^5 + 0.20 \mathbf{1}[\text{urgent}] + 0.25 \mathbf{1}[\text{new beneficiary}] + 0.20 \mathbf{1}[\text{privileged}] \\ 0 & \text{otherwise} \end{cases}}_{\text{transaction value}} \right). \quad (32)$$

7.2 Decision risk

$$R_d = 1 - (1 - R_a) \left(1 - \frac{1}{2}R_c\right) \quad (33)$$

Why this choice: a noisy-OR, and why context is halved

Reading R_a and R_c as independent probabilities that *something is wrong*, $1 - R_d = (1 - R_a)(1 - \frac{1}{2}R_c)$ is the probability that neither concern fires. This is the noisy-OR combination: it is monotone in both arguments, saturates correctly at 1, and never lets a low context risk *reduce* the acoustic risk — since $R_d \geq R_a$ always, with equality when $R_c = 0$. The factor $\frac{1}{2}$ caps how much business metadata alone can contribute, because a large payment is not evidence of voice cloning. A wire transfer of INR 10^7 with pristine audio reaches only $R_d = 0.175$.

The separation invariant

R_a is a function of acoustic evidence *only*. Context can never alter it. The two combine at exactly one place, R_d , and a unit test asserts that changing context metadata leaves R_a bit-identical. Collapsing the three numbers into one would make it impossible for an operator to know whether a call was flagged because the voice was suspicious or merely because the payment was large.

7.3 Decision bands

With warning threshold $\theta_w = 0.50$ and alert threshold $\theta_a = 0.75$:

$$\text{state} = \begin{cases} \text{ANALYZING} & R_a = \text{null} \\ \text{CRITICAL/HIGH} & R_d \geq \theta_a \\ \text{REVIEW} & R_d \geq \theta_w \text{ or } R_c \geq 0.7 \\ \text{LOW} & \text{otherwise.} \end{cases} \quad (34)$$

A publishable risk additionally requires $T_{\text{voiced}} \geq 4\text{s}$; otherwise the state is ANALYZING with reason LOW_EVIDENCE. Alerts *latch* until an independent verification is recorded.

Why this choice: latching, and never publishing a green state early

Risk is non-monotone: the age schedule can lower a score after a genuine alarm, so an unlatched policy would silently release a hold. Latching makes the safe action sticky until a human verification clears it. Similarly, the evidence gate ensures silence or a cold start reports ANALYZING rather than LOW — an unknown is never displayed as a clean bill of health.

8 Scheduling and latency

8.1 Multi-rate branch cadence

Branch i with cadence C_i runs at window k (audio time t_k) iff $t_k - t_{\text{last},i} \geq C_i$. A published result expires after $3C_i$; between runs the cached result is reused. Results are in one of three states — missing, fresh, or stale — and the latter two are never conflated with a zero score.

Why this choice: cadence on the audio clock, not the wall clock

If cadence were measured in wall-clock time, an accelerated offline replay (which processes 40 s of audio in 0.25 s) would find every branch “not yet due” and skip nearly all of them, making benchmarks meaningless and irreproducible. Measuring on the audio timeline makes an offline replay schedule branches exactly as a live call would, so evaluation and production agree by construction.

Measured effect at $W = 2$, $H = 0.5$: the artifact branch runs 39 times instead of 77 while all 77 windows still emit a fused score, giving a $1.63\times$ – $1.67\times$ throughput gain reproduced on two different processors.

8.2 Capture backpressure

The capture queue holds at most $N_q = 8$ completed windows. On overflow the *oldest* window is discarded and counted.

Why this choice: drop oldest, not newest

A live risk score is only useful if it describes the audio the caller is speaking now. Discarding the newest window would freeze the operator’s view in the past while the backlog drains — the precise failure the score exists to prevent. Dropping the oldest bounds latency at $N_q H$ seconds and makes the loss explicit rather than letting delay grow without limit. A drop also resets the continuity state, so the gap is not mis-scored as a splice.

8.3 Percentiles and real-time factor

The nearest-rank percentile of a sorted sample $z_{(1)} \leq \dots \leq z_{(n)}$ is

$$P_p = z_{(\lceil pn/100 \rceil)}, \quad \text{RTF} = \frac{\text{audio seconds processed}}{\text{compute seconds}}. \quad (35)$$

Why this choice: nearest-rank, and why p_{95} rather than the mean

Nearest-rank returns an *observed* value, never an interpolation between two measurements that were never made. Latency distributions are right-skewed, so the mean hides the tail that actually breaks a real-time budget; p_{95} against the hop H is the meaningful pass/fail test. $\text{RTF} > 1$ is the condition for keeping up with live audio. An empty sample returns `null`, not 0, since 0 ms would read as “instantaneous”.

9 Evaluation metrics

Let scores $\{z_i\}$ and labels $\{y_i\}$, $y_i = 1$ for spoof. For threshold θ ,

$$\text{FPR}(\theta) = \frac{|\{i : y_i = 0, z_i \geq \theta\}|}{|\{i : y_i = 0\}|}, \quad \text{TPR}(\theta) = \frac{|\{i : y_i = 1, z_i \geq \theta\}|}{|\{i : y_i = 1\}|}. \quad (36)$$

Equal error rate. With $\text{FNR} = 1 - \text{TPR}$, the EER is the value where $\text{FPR} = \text{FNR}$. Since the empirical ROC is a step function the crossing generally falls between two operating points a, b ; the implementation interpolates linearly on $g(\theta) = \text{FPR} - \text{FNR}$:

$$\lambda = \frac{-g_a}{g_b - g_a}, \quad \text{EER} = \text{FPR}_a + \lambda (\text{FPR}_b - \text{FPR}_a). \quad (37)$$

ROC-AUC. $AUC = \Pr(z_i > z_j \mid y_i = 1, y_j = 0)$, the probability that a random spoof outranks a random bonafide — equivalently the normalised Mann–Whitney U statistic. It is threshold-free.

Brier score and calibration error.

$$BS = \frac{1}{n} \sum_i (z_i - y_i)^2, \quad ECE = \sum_{b=1}^{10} \frac{|B_b|}{n} |\bar{z}_{B_b} - \bar{y}_{B_b}|. \quad (38)$$

Why this choice: reporting a family of metrics, not a single headline

EER assumes the two error types are equally costly, which is false here: a false positive costs a callback, a false negative costs a fraudulent transfer. AUC is threshold-free but insensitive to calibration. Brier and ECE ask whether a score of 0.8 actually means 80%, which matters because the policy engine thresholds on the number. Together they describe ranking quality, operating-point behaviour, and calibration — no one of them is sufficient.

Limitation: low-FPR operating points need cohort size

With n_0 genuine clips the finest resolvable false-positive rate is $1/n_0$. The measured cohort has $n_0 = 65$, so $1/65 \approx 1.5\%$ and a “TPR at 1% FPR” figure is *not resolvable*. The tooling emits `low_fpr_sample_warning` when $n_0 < 100$ and that number must not be quoted.

10 Measured results and findings

Corpus: ASVspoof 2019 LA evaluation partition (ODC-BY), 150 bonafide / 150 spoof, speaker-disjoint 31/31 split, reference index built from clean audio only. Extractor: `dsp-surrogate-v1`. Full report in `evidence/codec-benchmark.json`.

Condition	EER %	Δ EER	AUC	Δ AUC	Bona. med.	Spoof med.
clean	37.01	—	0.707	—	0.350	0.463
opus_16k	37.43	+0.42	0.688	−0.019	0.400	0.475
opus_32k	39.66	+2.66	0.686	−0.021	0.400	0.500
narrowband_8k	37.40	+0.40	0.676	−0.031	0.275	0.325
g711_ulaw	37.49	+0.48	0.676	−0.031	0.275	0.325
noise_10db	42.14	+5.13	0.639	−0.068	0.450	0.450

Finding 1 — noise dominates codec compression. Every codec leg costs only 0.019–0.031 AUC. Additive noise at 10 dB SNR costs 0.068, more than twice the worst codec, and collapses the median separation between bonafide and spoof to *exactly zero*. For this feature set the dominant robustness threat is noise, not compression.

Finding 2 — noise inflates perceived evidence while destroying it. Clips clearing the live-call evidence gate: 5 (clean) versus 28 (noise_10db), a $5.6\times$ increase. The mechanism is the energy activity gate of §2.3, whose voiced fraction rises from 0.564 to 1.000 under the same condition. The system becomes more confident it has sufficient evidence exactly when that evidence is least discriminative. Since SNR_{dB} already reads 19.6 dB and Q is available per window, the fusion has the signal required to discount this evidence; wiring Q into the fusion weights is the outstanding work.

Interpretation

AUC = 0.707 is weak but non-trivial signal from an unsupervised DSP representation on real spoofed speech. It is *not* a usable detector and not a deepfake verdict. Its purpose is to establish a measured floor on a real corpus, so that a trained countermeasure can be compared against something rather than asserted. No value in this document was estimated, extrapolated, or tuned to look better.

11 Summary of design principles

1. **Absent evidence is masked, never zeroed.** Missing, stale and failed branches are excluded from a renormalised weighted sum instead of contributing a score that would read as “genuine”.
2. **Unavailable measurements are null.** A quantity that could not be computed is never reported as 0, because 0 is a legitimate value for most of these quantities and would be indistinguishable from a real measurement.
3. **Channel reliability is orthogonal to risk.** Q scales how much the acoustic evidence is trusted; it is never added to the risk.
4. **Authenticity and context never mix except at the decision.** Enforced by a unit test.
5. **Bounded ranges throughout.** Every track score, penalty and risk is clipped to $[0, 1]$ so weights carry consistent meaning.
6. **Scale-free comparisons.** Log ratios, cosine similarity, and dimensionless ratios (jitter, shimmer, flatness) so gain and pitch do not act as confounds.
7. **Counterexamples are pinned, not tuned.** The EMA warm-up and mature-call dilution weaknesses are asserted by tests so they cannot be silently “fixed” by moving a threshold.

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